GG–C–466d
March 2, 1694

SUPERSEDING
Int. Fed. Spec. GG-C-00466c (GSA-FSS)
October 17, 1960 and
Fed. Spec. GG-C-466b
November 3, 1952.

FEDERAL SPECIFICATION

CLOCKS, WALL, (SYNCHRONOUS MOTOR)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers synchronous, self-starting motor wall clocks without current failure standby, for use on alternating current only.

1.2 Classification.

1.2.1 *Type and grade.* Clocks furnished under this specification shall be of one type and one grade.

1.2.2 *Sizes.* The synchronous motor wall clock dials shall be of the following sizes, as specified (see 6.1):

Size 1—8 inches
Size 2—12 inches
Size 3—15 inches

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications and Standards. The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

L-P-310—Phenolic Molded Plastics.
PPP-T-360—Time Measuring Instruments: Packaging of.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, and Auburn, Wash.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications, Standards, and Handbooks from established distribution points in their agencies.)

Military Standard:

MIL-STD-105 — Sampling Procedures and Tables for Inspection by Attributes.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification, and unless otherwise indicated, the issues in effect on date of invitation for bids shall apply.

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Underwriters' Laboratories, Inc., Publication:

Subject No. 863—Standard for Time Indicating and Recording Appliances.

(Copies may be obtained from Underwriters' Laboratories, Inc., Walt Whitman Road, Melville, N. Y. 11749; 207 E. Ohio Street, Chicago, Illinois, 60611; and 1655 Scott Blvd., Santa Clara, California 95050.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 Each bidder shall submit to the contracting officer proof that the complete electric wall clock unit he proposes to supply under this specification, conforms to all applicable requirements of the Underwriters' Laboratories, Inc., Standards, including UL 863, Standard for Time—Indicating and Recording Appliance. The label or reexamination listing of the Underwriters' Laboratories, Inc., may be accepted as evidence that the electric wall clock unit conforms to these requirements.

3.1.2 In lieu of the label or reexamination listing, the bidder may submit independent proof satisfactory to the contracting agency that the electric wall clock unit conforms to the requirements, including methods of tests, of Underwriters' Laboratories Standards.

3.1.3 Compliance with the above preliminary requirements regarding fire and casualty hazards, does not absolve the bidder from complete compliance with the other requirements of this specification in order to secure the acceptance of his complete electric wall clock unit.

3.2 Cases. Cases shall be of the surface type, round and suitable for hanging on concealed hangers. They shall be made of steel or a molded plastic material. The steel used shall be of 0.025 minimum thickness. Steel cases shall be finished with acid phosphate dip and sprayed with a zinc-chromate undercoating before application of final finish. The molded plastic material shall be in accordance with any of the cellulosic types II through VI of Federal Specification L-P-301. Unless otherwise specified, metal cases shall be finished in bronze, dark brown or gray lacquer or enamel which will not be easily chipped or cracked. Molded plastic cases shall be supplied in standard mold finish. All cases shall have convex cover glasses. All cases shall have openings for the cord, placed in a concealed or inconspicuous location, smoothly rounded to prevent wear on the cord.

3.3 Dial. Unless otherwise specified, see 3.3.2, 12-hour dials shall be furnished. The diameter of the dials (sight opening), shall be of the dimensions indicated in 1.2.1, plus or minus 5 percent. Dials shall have black or blue-black hands and black arabic numerals and minute marks against a black background and shall be treated to prevent discoloration. When the dial is of material other than metal it shall be securely fastened to a metal base in such a manner as to prevent warping, buckling or any irregularities in appearance or positive fit.

3.3.1 Dials shall have sweep second hands in addition to hour and minute hands. Second hand shall be finished in some distinguishing color or shade.

3.3.2 When specified in the invitation for bids, dials shall register 24 hours in one revolution of the hour hand, 1 hour in one revolution of the minute hand and 1 minute in one revolution of the second hand. The figures indicating the hours shall be arabic numerals of two distinct sizes, the figures "00", "4", "8", "12", "16" and "20" being approximately 50 percent higher than the remaining figures. The figures indicating the minutes and seconds shall be not less than 6/32 inch high for 12-inch clocks and proportionately larger or smaller for the larger and smaller sizes of clocks.

3.4 Movements.

3.4.1 *Mechanical.* All movements shall be first grade both as to material and work-

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manship. Gears shall be of clock brass, or of steel, nylon, aluminum alloy or laminated phenolic material of equal or superior wearing quality. Fiber gears will not be acceptable. All pivots shall be smoothly finished and free from burrs. All gears and pinions shall be cleancut and free from burrs. Gears and pinions shall have hubs or equivalent construction necessary to insure rigidity and shall be force fit on the staffs. All shafts and gears which operate above six revolutions per minute shall be contained in a unit assembly in which special provisions shall be made for the self lubrication of all bearings and gears through the use of an efficient capillary or other equally effective feed lubrication system and a sealed-in supply of high quality clock oil adequate to assure trouble free operation of the sealed unit for at least 5 years. The design of the unit assembly and the characteristics of the oil sealed in it shall be such that at least 75 percent of the initial oil supply will be available after two years of use. The lubricant used in each sealed unit shall be rated as to its physical and chemical characteristics including those listed in table I. The methods used to measure and evaluate these characteristics shall be those indicated in table I. (See 6.3.)

TABLE I

Characteristic	Method of evaluating
Evaporation loss	ASTME D 972-56
Color of oil	ASTME D 1500-60T
Viscosity (Kinematic)	ASTME D 445-53T
Precipitation number	ASTME D 91-52
Flash and fire point	ASTME D 92-57
Corrosion and oxidation stability	ASTME D 5308.4
Cloud and pour points	ASTME D 97-57
Acid number	ASTME D 974-58T

3.4.1.1 Characteristics of the oil shall be such as to insure synchronous operations at all temperatures from 10° to 125°F. All mechanism shall be mounted within the clock case and shall operate without objectionable noise or hum. Parts subject to wear or replacement shall be easily removed and installed and shall be interchangeable in clocks of the same manufacturer and size. The overall design shall be such that the sealed unit referred to above can be easily replaced in five minutes or less. Its replacement shall require no tools beyond a screwdriver of appropriate size.

3.4.2 The devices for resetting shall project from the bottom of the case. The resetting device shall be of the pull-down and reset type. The push-up type will not be acceptable.

3.4.3 *Electrical.* All electrical components shall be so designed and constructed as to comply with Underwriters' Laboratories, Inc., Standard UL No. 863, Time-Indicating and Recording Applicances.

3.4.3.1 Unless otherwise specified, the motor shall be of the synchronous self-starting type for use on 110 to 120 volts, 60 cycle, alternating current. (See 6.1.)

3.4.3.2 The electric motor used shall be fully capable of starting the clock and driving its movement while maintaining synchronous speed when connected to a 60 cycle 90 volt (Approx. 80 % of full voltage) source of power. When connected to 120 volts, 60 cycles, the power required to operate clocks up to and including 16-inch dial clocks shall not exceed 3 watts per clock.

3.4.3.3 A cord having an exposed length of not less than 5 feet, measured from the outside of the back cover with a molded-on type, rubber, neoprene, or thermoplastic insulated, parallel blade attachment-plug-cap shall be provided with the clock. The cord shall be type X, XT, SP-1, or SPT-1, integral construction, or shall be of a type having at least equivalent properties.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Supplier responsibilities for inspection. The supplier is responsible for the perform-

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ance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract or order (see 6.2). The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Inspection.* Inspection as used in this specification, is defined as both examination (such as visual, auditory, etc., investigation without the use of special laboratory appliances or procedures) and testing determination by technical means of physical and chemical properties of the item.

4.1.2 *Certification of quality.* When available, certificates of quality listing the specified test method and test results obtained, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor.

4.2 **Sampling.** Sampling shall be conducted in accordance with Military Standard MIL-STD-105 (see table II of this specification).

4.2.1 *Classification of defects.* For inspection purposes, defects shall be classified as follows:

Examine	Defect	Major	Minor
Case	Material used other than specified	X	
	Surface not smooth	X	
	Opening in case for chord not smoothly rounded	X	
	Exposed length of chord less than 5 feet		X
	Improper fit of case to dial and cover glass	X	
Clock-General	Not design specified	X	
	Case improperly finished	X	
	Burrs or undesirable projecting ridges at the dividing line of mold		X
	Any component cracked, fractured, broken or damaged	X	
	Clock will not run	X	
	Clock operates with objectionable noise or hum	X	
	Sealed unit cannot be easily replaced with simple tools	X	
Glass	Cover glass broken	X	
	Glass poor quality, not clear, blisters, bubbles, scratch or scratches, coarse or open seeds, heavy seeds or feathers, or other defects which distort or impair the vision		X
Dial	Any dial marking missing, illegible, incomplete, incorrect, or wrong material used to apply	X	
	Dial surface scratched, abraded, or otherwise damaged	X	
Hands	Any hand not securely mounted	X	
	Any hand bent, malformed, or otherwise damaged	X	
	Hand rubs against glass or dial, or hand obstructs movements of another hand	X	
	Second hand not of distinguishing color		X

4.3 **Examination.** The clocks shall be examined to determined compliance with all requirements contained in sections 3 and 5 of this specification. Non-conformance with these requirements will be permitted to the extent indicated in 4.2.

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TABLE II

For visual examination	Inspection level	AQL (percent defective)	Minimum sample size
Major	II	1.0	—
Total	II	4.0	—
For testing	L4	*	2

* Acceptance number shall be zero (0) for all sample sizes.

4.4 **Test.**

4.4.1 *Synchronous operation.* The source of power to be used in this test shall be 90 volt, 60 cycles, and of sufficient capacity such that the simultaneous connecting of the clocks to be tested will have a negligible effect on the voltage at its terminals. Clocks to be tested shall be identified by number and shall be hung on a vertical surface in the normal running position, in a room maintained at 70° ± 5°F. for at least 24 hours prior to making the test. All clocks to be tested shall be plugged into multiple outlets fed by a common supply cord arranged so that it may conveniently be connected to the 90-volt source of power. Two control clocks when connected to 90 volts, 60 cycles current and known to keep time correct to within one second in 96 hours when compared with standard time signals controlled by the National Bureau of Standards, shall also be connected to the common supply cord. Care shall be taken to see that all connections are positive. The methods of determining and establishing the accuracy of the control clocks to within the required one second in 96 hours is within the discretion of the manufacturer, but must be entirely adequate for the purpose. The following sequence shall be followed in making the test:

(1) The power supply source shall be adjusted to 90 volts, plus or minus one volt, and the common supply cord shall be connected to the source.

(2) Clocks which do not start, i.e., second hand start to rotate, within ten seconds, shall be considered to have failed and need not be considered further in this test.

(3) The time indicated by all clocks still under test, shall be read simultaneously or in a definite time sequence, and recorded.

(4) At the end of 24 hours, plus or minus 1 hour, the time indicated by all clocks still under test, shall be again read simultaneously or in the same time sequence as used in making the initial readings. All clocks for which the test show a difference in readings of more than three seconds for the test period when compared to the two control clocks, shall be considered to have failed.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. Packaging, packing, and marking shall be as specified for packaging Group 4 in Federal Specification PPP-T-360.

6. NOTES

6.1 **Ordering data.** Purchasers should exercise any desired options offered herein and procurement documents should specify the following:

(*a*) Title, symbol, and date of this specification.

(*b*) Size of dial and case material and finish (see 1.2.1 and 3.2).

(*c*) Dial if other than as specified (see 3.3 and 3.3.2).

(*d*) Voltage and frequency of operation (see 3.4.3.1).

(*e*) Levels of packaging, packing and marking for shipment.

(*f*) How and when record of tests and required data is to be made available to Government.

6.2 GUARANTEE: The clocks furnished under this invitation for bid shall be and are guaranteed against mechanical defects for a period of one year and all necessary repairs or adjustments will be made and any defective or broken parts not caused by mis-

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handling or accident (except accident of other than the Government) will be promptly replaced by the contractor during the guarantee period free of any cost to the Government with all replacement costs paid by the contractor including transportation within the continental United States.

6.3 The right is reserved to reject bids on products that have not been subjected to the required tests (4.4), or for which the supplier is unable to furnish authenticated records (3.4.1) upon request of the Government to satisfy the contracting agency that clocks of like design and construction have met all requirements of this specification.

6.4 This specification does not include all types, classes, grades, sizes, etc., of the commodity indicated by the title of the specification, or which are commercially available, but is intended to cover the types, etc., which are normally procured to meet Federal Government requirements.

6.5 It is believed that this specification adequately describes the characteristics necessary to secure the desired material, and that normally no samples will be necessary prior to award to determine compliance with this specification. If, for any particular purpose, samples with bids are necessary, they should be specifically asked for in the invitation for bids and the particular purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.6 **Transportation description.** Transportation description and minimum weights applicable to this commodity are:

Clocks, NOIBN

Carload minimum weight 16,000 pounds, subject to Rule 34, Uniform Freight Classification

Clocks, NOI

Truckload minimum weight (w) 16.6 pounds, subject to Rule 115, National Motor Freight Classification.

Notice. When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise, as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:

Army—O Sig

Navy—Mc Sh Y

Air Force

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